

Project 1

Data Cleaning and Preparation The Dataset contained 14 Column Headers, 1,201 Rows (including headers), and 309 Blank Cells. Data Cleaning was performed using Microsoft Excel Power Query.

Steps Taken to Perform Data Cleaning

STEP 1: Converted the Main Dataset into a Table

- Shortcuts: Ctrl + A to select the entire dataset range.
- Ctrl + T to convert the range into an official Excel Table.

STEP 2: Exported the Table into Power Query

- Navigated to **Data Tab** → Clicked **From Sheet / Table/Range** to load the dataset into the Power Query Editor.

STEP 3: Converted Excel Serial Numbers to Standard ISO Dates

- Selected the Date column and cleared conversion errors.
- Used custom M syntax (Date.ToText(Date.From(Number.From([Date])), "yyyy-MM-dd")) to convert numeric serials (e.g., 44930) into standard ISO 8601 (YYYY-MM-DD) format.

STEP 4: Adjusted and Standardized Data Types

- **OrderID**: Text
- **Date**: Date
- **CustomerID**: Text
- **Product**: Text
- **Quantity**: Whole Number
- **UnitPrice**: Currency / Fixed Decimal
- **ShippingAddress**: Text
- **PaymentMethod**: Text
- **OrderStatus**: Text
- **TrackingNumber**: Text
- **ItemsInCart**: Whole Number
- **CouponCode**: Text
- **ReferralSource**: Text
- **TotalPrice**: Currency / Fixed Decimal

STEP 5: Handled Blank Cells (Strategic Imputation)

- The CouponCode column contained 309 blank/null cells.
- Replaced all null values with "NO_COUPON" to avoid listwise deletion and preserve full row context.

STEP 6: Standardized Text Fields

- Applied **Trim** and **Capitalize Each Word** (Proper Case) across Product, PaymentMethod, OrderStatus, and ReferralSource to eliminate irregular spacing.

STEP 7: Performed Integrity Audit and Duplicate Check

- Applied **Remove Duplicates** on the OrderID primary key column.
- Confirmed 0 duplicate records across all 1,200 unique order IDs.

STEP 8: Reordered and Positioned Columns

- Moved the cleaned Date column directly next to OrderID for logical table flow.

STEP 9: Exported Transformed Data Back into Excel

- Navigated to **Home Tab** → Clicked **Close & Load** to output the clean table into a new worksheet.

STEP 10: Formatted Financial Columns in Worksheet

- Highlighted UnitPrice and TotalPrice columns and pressed Ctrl + 1.
- Applied **Number** format set strictly to **2 Decimal Places** to ensure uniform display (e.g., 239.00, 2853.10).

STEP 11: Renamed Transformed Data Table

- Renamed the output table and sheet to Cleaned_Data.

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